NATIONAL INSTITUTE OF TECHNOLOGY PATNA

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Mid Semester Examination, October 2024

Course Code: CH14101 (CSE-I) **Time: 2.00 Hours**

Course Title: Engineering Chemistry **Full Marks: 30**

Answer all questions

Q1. (a) Explain with example primary and secondary fuels. Write the characteristics of a good fuel.

(b) Write the process for the determination of nitrogen in a fuel sample.

(c) What is Heavy oil? Write its different fractions and their application.

(d) What is sweetening of petrol?

(CO1) 3+3+3+1= 10

Q2. (a) The ultimate analysis of a coal gave the following results: C = 84%, S = 1.5%, N = 0.6%, H = 5.5%, and O =8.4%. Calculate the gross and net calorific value of the coal using Dulong's formula.

(b) Explain fuse wire correction, acid correction and cooling correction in the determination of calorific value by bomb calorimeter.

(c) Calorific value of 1.5 g of a coal sample was determined in a bomb calorimeter. The ash formed was extracted with acid and the acid extract heated with $BaCl_2$ to obtain a precipitate of $BaSO_4$. The precipitate was filtered, washed and found to weigh 0.55g. Calculate the percentage of sulphur in the coal sample.

(d) What is cracking?

(CO1) 3+3+3+1=10

Q3. (a) Explain Kohlrausch law and its application.

(b) Write the effect of temperature and concentration on conductance and explain with graph..

(c) A 0.01 N KCl solution shows a resistance 225 ohms in a conductivity cell. The specific conductivity of 0.01 N KCl solution at the temperature of experiment is 0.00141 mho/cm. If a 0.02 N solution of an acid shows a resistance of 80 ohms in the same cell, find the specific and equivalent conductance of the acid.

(d) Write the expression of degree of dissociation in terms of conductance.

(CO1) 3+3+3+1=10

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